Megha Srivastava

megha@cs.stanford.edu • meghabyte.github.io
735 Campus Dr., Stanford, CA 94305

I work at the intersection of **machine learning**, **safety**, and **security**, focusing on **reducing computational and cognitive costs of verifying modern AI systems**. Beyond shaping model behavior through training-time objectives, my research treats verification as an end-to-end problem, spanning hardware-level reproducibility on the computational side and models of human learning under AI assistance on the cognitive side. With applications in software development, robotics, and small-molecule drug discovery, my work has received an ICML Best Paper Runner-Up Award, been supported by the NSF GRFP and IBM PhD Fellowships, and appeared in venues including NeurIPS, ICML, and Nature.

Education

Stanford University • Stanford, CA — June 2027 *(expected)*
PhD in Computer Science, co-advised by Dorsa Sadigh & Dan Boneh
Supported by the NSF Graduate Research Fellowship (2018 – 2023) and IBM PhD Fellowship (2023-2024)

Massachusetts Institute of Technology • Cambridge, MA — Fall 2023 – Winter 2024
Visiting PhD student at the Computer Science & Artificial Intelligence Laboratory, hosted by Jacob Andreas

Stanford University • Stanford, CA — Spring 2019
BS+MS Coterminal Degree in Computer Science with Honors and Distinction, Minor in Creative Writing
Research advised by Percy Liang & Tatsunori Hashimoto
Ben Wegbreit Prize for Best Undergraduate Honors Thesis in Computer Science
Bing Overseas Studies Program at Oxford University (Logic & Computability) in Spring 2017

Experience

Research Intern – Genesis Molecular AI — Summer 2026
- Deep learning research at the intersection of generative AI and drug discovery.
- Led to *"Pyrite: A Benchmark for Protein-Ligand Pose Ranking Agents"* (NeurIPS 2026 AgenticLS Workshop).

Member of Technical Staff (Contractor) – Anthropic — Spring 2025 – present
- Supported technical writing and research mentorship of AI Fellows (e.g., Findings from a Pilot Anthropic—OpenAI Alignment Evaluation Exercise 🔗).

AI Resident – Microsoft Research — Fall 2019 – Fall 2020
- Quantifying backward compatibility impact of retraining ML models (Mentors: Besmira Nushi, Eric Horvitz).
- Led to *"An Empirical Analysis of Backward Compatibility in Machine Learning Systems"* (KDD 2020).

Research Intern – Google Research — Summer 2019
- Improving robustness of image understanding models with the Research & Machine Intelligence team.

ETH Zürich Student Summer Research Fellowship – Learning & Adaptive Systems Group — Summer 2018
- Active learning and adaptive hypothesis testing (Mentors: Hoda Heidari, Andreas Krause)
- Led to *"Mathematical Notions vs. Human Perception of Fairness: A Descriptive Approach to Fairness for Machine Learning"* (KDD 2019).

Research Assistant – Vision & Perception Neuroscience Group — Summer 2016
- Generalization and perceptual invariances in both human and artificial vision models (Mentor: Kalanit Grill-Spector)
- Supported by the Bio-X Undergraduate Research Award.

Under Submission

Megha Srivastava, Michael H. Lim, Allison M. Keys, Nate Gruver, Pranav Murugan. *"Pyrite: Building and Benchmarking Protein-Ligand Pose Ranking Agents."* Under submission to Transactions on Machine Learning (TMLR). Preliminary version at the NeurIPS Agentic AI for Biological Discovery Workshop, 2026.
AI for Science·Robust ML

Megha Srivastava, Jonathan Ouyang, Eric Zhou, Andrew Silva, Emily Sumner, Dorsa Sadigh, Yuchen Cui, Deepak Gopinath, Guy Rosman. *"Proximal State Nudging: Reducing Skill Atrophy from AI Assistance."* Under submission to IEEE Robotics and Automation Letters (RA-L). Preliminary version at the ICML Trustworthy AI for Good Workshop, 2026.
AI for Education·Embodied AI

Publications (Google Scholar, * means equal contribution)

Erez Badash, Dan Boneh, Ilan Komargodski, **Megha Srivastava**. *"Hawkeye: Reproducing GPU-Level Non-Determinism."* Ninth Annual Conference on Machine Learning and Systems (MLSys), 2026. (alphabetical author ordering)
Robust ML·Security

Alexis Ross*, **Megha Srivastava***, Jeremiah Blanchard, Jacob Andreas. *"Modeling Student Learning with 3.8 Million Program Traces."* International Conference on AI in Education (AIED), 2026.
AI for Education·NLP

Publications Cont. (Google Scholar, * denotes equal contribution)

Jan Betley*, Niels Warncke*, Anna Sztyber-Betley, Daniel Tan, Xuchan Bao, Martín Soto, **Megha Srivastava**, Nathan Labenz, & Owain Evans *"Training Large Language Models on Narrow Tasks Can Lead to Broad Misalignment."* Nature, 2026.
Robust ML·NLP

Megha Srivastava, Cédric Colas, Dorsa Sadigh, Jacob Andreas. *"Policy Learning with a Language Bottleneck."* Transactions on Machine Learning (TMLR), 2026.
NLP·Embodied AI ·Human-AI Interaction

Megha Srivastava*, Reihaneh Iranmanesh*, Yuchen Cui, Deepak Gopinath, Emily Sumner, Andrew Silva, Laporsha Dees, Guy Rosman, Dorsa Sadigh. *"Shared Autonomy for Proximal Teaching."* ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2025.
AI for Education ·Embodied AI

Megha Srivastava, Simran Arora, Dan Boneh. *"Optimistic Verifiable Training by Controlling Hardware Nondeterminism."* Advances in Neural Information Processing Systems, 37 (NeurIPS), 2024.
Robust ML·Security

Neil Perry*, **Megha Srivastava***, Deepak Kumar, Dan Boneh. *"Do Users Write More Insecure Code with AI Assistants?"* ACM Conference on Computer and Communications Security (CCS), 2023.
mlsec.org Top-100 Computer Security Papers 🔗
Security ·Human-AI Interaction

Mina Lee, **Megha Srivastava**, Amelia Hardy, John Thickstun, Esin Durmus, Ashwin Paranjape, Ines Gerard-Ursin, Xiang Lisa Li, Faisal Ladhak, Frieda Rong, Rose E. Wang, Minae Kwon, Joon Sung Park, Hancheng Cao, Tony Lee, Rishi Bommasani, Michael Bernstein, Percy Liang *"Evaluating Human-Language Model Interaction."* Transactions on Machine Learning Research (TMLR), 2023.

Led project experimental design; follow-up analysis and dataset release with John Thickstun (🔗)

§ **Question Answering**: Megha Srivastava, John Thickstun, Rose Wang, Minae Kwon, Mina Lee
§ **Crossword Puzzles**: Megha Srivastava

Human-AI Interaction ·NLP

Megha Srivastava, Noah Goodman, Dorsa Sadigh. *"Generating Language Corrections for Teaching Physical Control Tasks."* Proceedings of the 40th International Conference on Machine Learning (ICML), 2023.
AI for Education ·Embodied AI

Megha Srivastava, Erdem Biyik, Suvir Mirchandani, Noah Goodman, Dorsa Sadigh. *"Assistive Teaching of Motor Control Tasks to Humans."* Advances in Neural Information Processing Systems, 35 (NeurIPS), 2022.
AI for Education ·Embodied AI

Siddharth Karamcheti*, **Megha Srivastava***, Percy Liang, Dorsa Sadigh. *"LILA: Language-Informed Latent Actions."* Conference on Robot Learning (CoRL), 2021.
Embodied AI·NLP

Megha Srivastava and Noah Goodman. *"Question Generation for Adaptive Education."* Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL), 2021.
AI for Education·NLP

Megha Srivastava, Tatsunori Hashimoto, Percy Liang. *"Robustness to Spurious Correlations via Human Annotations."* Proceedings of the 37th International Conference on Machine Learning (ICML), 2020.
Robust ML·NLP·Human-AI Interaction

Megha Srivastava, Besmira Nushi, Ece Kamar, Shital Shah, Eric Horvitz *"An Empirical Analysis of Backward Compatibility in Machine Learning Systems."* Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD), 2020.
Robust ML

Megha Srivastava, Hoda Heidari, Andreas Krause. *"Mathematical Notions vs. Human Perception of Fairness: A Descriptive Approach to Fairness for Machine Learning."*, Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD), 2019.
Human-AI Interaction

Tatsunori Hashimoto, **Megha Srivastava**, Hongseok Namkoong, Percy Liang. *"Fairness Without Demographics in Repeated Loss Minimization,"* Proceedings of the 35th International Conference on Machine Learning (ICML), 2018.
Best Paper Runner-Up Award
Robust-ML

Other Works (Blog Posts, Abstracts, Technical Reports, etc.)

- **Megha Srivastava** *"Echo: A multi-agent AI system for patient-centered pharmacovigilance."* 1st Open Conference of AI Agents for Science, 2025. **Spotlight Talk**. *[Experimental conference exploring human-AI collaboration in research. More details in blog post on research process 🔗]*
- Sam Bowman, **Megha Srivastava**, Jon Kutasov, Rowan Wang, Trenton Bricken, Benjamin Wright, Ethan Perez, and Nicholas Carlini. *"Findings from a Pilot Anthropic—OpenAI Alignment Evaluation Exercise."* Anthropic, 2025. 🔗
- Rose Wang and **Megha Srivastava**. *"Productive Struggle: The Future of Human Learning in the Age of AI."* The Stanford AI Lab Blog, 2025. Featured and translated in Spanish by the Inter-American Development Bank. 🔗
- **Megha Srivastava** and John Thickstun. *"Observations from HALIE: A Closer Look at Human-LM Interactions in Information-Seeking Contexts."* Center for Research on Foundation Models Blog, 2023. 🔗
- Kristin Lauter$^{\alpha\beta}$, Cathy Yuanchen Li$^{\alpha\beta}$, Krystal Maughan$^{\alpha\beta}$, Rachel Newton$^{\alpha\beta}$, **Megha Srivastava**$^{\alpha\beta}$. *"Machine learning for modular multiplication."* Proceedings of the WIN6 Workshop at Banff International Research Station for Mathematical Innovation and Discovery, 2023.
- **Megha Srivastava**, David Remus, Kalanit Grill-Spector. *"The Role of Learning in Complex Object Recognition and Discrimination Across Spatial Transformations: An Experimental Comparison of Artificial CNNs and Human Subjects."* Vision Sciences Society (VSS), 2017.

Selected Awards & Fellowships

- Human Robot Interaction (HRI) Pioneers, 2025
- Rising Star in Machine Learning (University of Maryland), 2023
- IBM PhD Fellowship, 2023
- American Association for the Advancement of Science (AAAS) Mass Media Fellowship (declined), 2019
- International Conference on Machine Learning (ICML) Best Paper Runner-Up Award, 2018
- National Science Foundation Graduate Research Fellowship, 2018
- Ben Wegbreit Prize for Best Undergraduate Honors Thesis in Computer Science (Stanford), 2018
- Phi Beta Kappa, 2018
- Tau Beta Pi, 2018
- Bio-X Research Award (Stanford), 2016
- Finalist, Lunsford Award for Oral Presentation of Research (Stanford), 2016

Teaching and Mentoring

- Course Assistant for (i) CS 224R (Deep Reinforcement Learning, 2026), CS 255 (Introduction to Cryptography, 2025), and CS 221 (Introduction to Artificial Intelligence, 2018). Mentored student group projects, taught discussion sections, performed grading, and held office hours.
- Mentored research of undergraduate and MS students resulting in several papers: *Reihaneh Iranmanesh (Amherst College → PhD at Georgetown University, co-author on HRI'25 paper), Zhiyin Lin (Stanford → OpenAI, CURIS program), Mallika Parulekar (Stanford), Jonathan Ouyang (UCLA), and Benita Wong (Stanford).*
- Instructor for Stanford Splash (Demystifying Hot Topics in Computer Science, 2017).

Service

- Science Small Groups Mentor (2024).
- Stanford AI Lab Blog Editor, 2020–present.
- Student Program Chair for Women in Machine Learning (WiML) 2023, co-located with NeurIPS 2023 .
- Reviewer for NeurIPS (2020 – present), ICLR (2021 – present), ICML* (2021 – present), RA-L 2022, CoRL 2023, L4DC 2024, CHI 2025, HRI-LBR 2026. (* recognized with ICML 2026 Gold Reviewer designation)

Guest Lectures

1. "Security of Modern Machine Learning Systems." *AI4All, June 2025*.
2. "Beyond Instruction Following: Language and Human-Robot Interaction." *Stanford University (Guest Lecture for CS 329X: Human-Centered NLP), October 2024*.
3. "Do Users Write More Insecure Code with AI Assistants?" *University of Maryland College Park (Guest Lecture for Large Language Models, Security, and Privacy seminar), October 2023*.

Research Talks

1. "Children of the Magenta Line: Skill Atrophy and Learning-Aware Shared Autonomy" *University of Washington Robotics Colloquium, November 2026.*
2. "Modeling Student Learning with 3.8 Million Program Traces" *Together AI, February 2026.*
3. "Modeling Student Learning with 3.8 Million Program Traces" *Deep Learning: Classics and Trends (ML Collective), November 2025.*
4. "Security of Modern Machine Learning" *Accenture Responsible AI Course, September 2025.*
5. "Optimistic Verifiable Training by Controlling Hardware Nondeterminism" *Google Privacy in ML Seminar, February 2025.*
6. "Security of Modern Machine Learning" *DaVita x Stanford Human-Centered AI, February 2025.*
7. "Optimistic Verifiable Training by Controlling Hardware Nondeterminism" *AI and Cryptography Workshop, Joint Mathematics Meeting, January 2025.*
8. "Optimistic Verifiable Training by Controlling Hardware Nondeterminism" *University of Washington Security Seminar, January 2025.*
9. "New Challenges of Trust with Large-Scale AI Systems" *University of Chicago, November 2024.*
10. "Security of Modern Machine Learning" *Stanford Cybersecurity & Privacy Festival, October 2024.*
11. "Policy Learning with a Language Bottleneck" ***Spotlight Talk*** *at Training Agents with Foundation Models Workshop (RLC 2024), August 2024.*
12. "Optimistic Verifiable Training by Controlling Hardware Nondeterminism" *University of Toronto, July 2024.*
13. "Security of Modern Machine Learning" *Accenture Responsible AI Course, July 2024.*
14. "Optimistic Verifiable Training by Controlling Hardware Nondeterminism" *Stanford Security Forum, April 2024.*
15. "Challenges in Human-AI Interaction for Information-Seeking Tasks" *Rising Stars in Machine Learning Workshop, November 2023.*
16. "Robustness to Spurious Correlations via Human Annotations" *Two Sigma PhD Symposium, June 2023.*
17. "Do Users Write More Insecure Code with AI Assistants?" *UC Berkeley Security Seminar, May 2023.*
18. "Assistive Teaching of Motor Control Tasks to Humans" *Stanford Collaborative Haptics in Robotics and Medicine Lab, May 2023.*
19. "Assistive Teaching of Motor Control Tasks to Humans" *SystemX Alliance Fall Conference, November 2022.*
20. "Assistive Teaching of Motor Control Tasks to Humans" *Simons Institute Workshop on AI & Humanity, July 2022.*
21. "Mathematical Notions vs. Human Perception of Fairness: A Descriptive Approach to Fairness for Machine Learning" *Oxford University Algorithms at Work Reading Group, May 2021.*
22. "Fairness & Robustness with Missing Information" *Stanford Causality & Cognition Lab, December 2020.*
23. "Mathematical Notions vs. Human Perception of Fairness: A Descriptive Approach to Fairness for Machine Learning" *Microsoft Research AI & Society Reading Group, October 2019.*